

Perception engineer with five years taking computer vision systems from research prototype to product on real vehicles. I own complete perception stacks, from sensor calibration and fusion to multi-target tracking, monocular depth estimation and VLM based scene understanding, delivered under hard real-time constraints.

01 EXPERIENCE

Five years shipping perception on real vehicles.

2nd

place at A2RL in the team's first eligible autonomous racing competition

250 km/h

closed-loop perception validated at race speed

4

perception and navigation sub projects owned in parallel

04 CORE SKILLS

PERCEPTION & ML

- Object detection & tracking
- Multi-target tracking (JIPDA)
- Monocular depth estimation
- Sensor calibration & fusion
- Vision-language models
- Synthetic data generation
- Model optimization

ENGINEERING

- C++ Python ROS
- Docker Git SQL
- MATLAB & Simulink

LANGUAGES

- Italian — native
- English — C2 French — B2
- Spanish — B1

02 EDUCATION

Milan, Rome and a semester in Paris.

Senior Perception Engineer · SIRB AI

Jan 2026 — Present · Abu Dhabi

- Perception lead for swarm autonomy, taking a research stack to a commercially deployed product.
- Architect the end-to-end perception framework for the research to product transition: JIPDA multi-target tracking, VMTI output, model optimization for embedded deployment, and a synthetic and real dataset pipeline for retraining.
 - Develop monocular depth and target position estimation from a single camera, with no depth sensor or LiDAR, and apply VLMs to classify detected entities and their affiliation.
 - Own 4 parallel sub projects across vision and navigation, one developed jointly with an external startup, and coordinate integration with the other module leads.
 - Design field demos jointly with the sub team leads, validate them on site and present them to clients.

Senior Perception Engineer · Technology Innovation Institute

Oct 2023 — Dec 2025 · Abu Dhabi

- Perception lead for the TII Car Racing team, racing a driverless car wheel-to-wheel above 250 km/h.
- Led the Perception Division to a 2nd place finish in the team's first eligible A2RL competition: designed, reviewed and integrated the startup built camera, radar and LiDAR pipeline (data collection, calibration, fusion, opponent tracking, visual localization error) inside a hard real-time latency budget.
 - Specified, reviewed and integrated an odometry based object speed estimation filter built by a 5+ engineer external startup team: set requirements, built the dataset, benchmarked and shipped on the car.
 - In the competition's earliest phase, selected and benchmarked the sensor suite and delivered it with a baseline perception stack to the organizers as technical consultant.

Customer Solutions Engineer · Unchained Labs

Dec 2024 — Sep 2025 · Remote

- Led product definition and delivery of Hunter, an AI agent for on-chain analytics, from scoping to release, alongside a full-time engineering role.

Research Assistant, Robotics & CV · Polytechnic of Milan

Sep 2022 — Jul 2023 · Milan

- Designed and deployed the perception stack for the Aida project, later spun off as Niulinx, an urban mobility autonomous driving startup.
- Collaborated with Maserati S.p.A. on 1000 Miglia 2023, fielding the first autonomous vehicle in the race's history.

Undergraduate Researcher, CV · IZONICS

Jan 2022 — Jun 2022 · Paris

- Built a 100+ sample dataset and trained a CNN to 95% accuracy on biological vs. PVC particle classification; reported results directly to the founders.

MSc Automation & Control Engineering

2020 — 2023 · Milan, Italy

Polytechnic of Milan
110 cum laude/110. Thesis nominated for the Award for Best Thesis on Advanced A&C Techniques.

Erasmus+ Exchange Programme

2022 · Paris, France

ESIEE Paris, Grande École d'Ingénieurs

BSc Mechanical Engineering

2016 — 2020 · Rome, Italy

La Sapienza, University of Rome

03 SELECTED PROJECTS

Two shipped on the road, two shipped on the web.

A2RL Autonomous Racing 2024–25
Camera, radar and LiDAR perception for a driverless race car. 2nd place, debut season

1000 Miglia 2023 2023
On a Maserati MC20 Cielo, the first autonomous vehicle to run the historic road race

Kymatics 2024
Voice to deployed web app code assistant, model agnostic. Built at Mistral Worldwide Hacks

Hunter 2024–
AI agent turning on-chain activity into readable insight; owned product and release